

Functional Dependencies

A functional dependency is a relationship between two sets of attributes in a relational database. It indicates that one attribute's value uniquely determines another's value.

Given a relation R, a functional dependency is denoted as:

$$X \rightarrow Y$$

which means X functionally determines Y or Y is determined by X.

Example:

Consider a table of Students with the following attributes:

- **StudentID**
- **Name**
- **Course**

A functional dependency might be: $\text{StudentID} \rightarrow \text{Name}$

This means that each StudentID uniquely determines a Name. If you know the StudentID, you can find out the Name.

Types of Functional Dependencies:

1. Trivial Functional Dependency:

- A functional dependency $X \rightarrow Y$ is trivial if Y is a subset of X .
- Example: $\{StudentID, Name\} \rightarrow StudentID$.

2. Non-Trivial Functional Dependency:

- A functional dependency $X \rightarrow Y$ is non-trivial if Y is not a subset of X .
- Example: $StudentID \rightarrow Name$.

Example:

Trivial FD-

Imagine we have a table Employee with the following attributes:

- **EmployeeID**
- **Name**

Now, consider the functional dependency: $\text{EmployeeID, Name} \rightarrow \text{Name}$

This FD is trivial because Name is already part of the left-hand side (i.e., EmployeeID, Name).

Non Trivial FD-

Consider the same Employee table with these attributes:

- **EmployeeID**
- **Name**
- **Department**

Now, look at this functional dependency: $\text{EmployeeID} \rightarrow \text{Name}$

This FD is non-trivial because the Name is not part of the left-hand side. Here, it means that each EmployeeID uniquely determines the Name

Properties of Functional dependencies

1. Reflexivity: If Y is a subset of X , $X \rightarrow Y$ i.e. X determines Y is a trivial functional dependency.

2. Augmentation : If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Ex. If $S_ID \rightarrow Sname$

then $S_ID, Course \rightarrow Sname, Course$

3. Transitive: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

Ex. If $S_ID \rightarrow Course$ and $Course \rightarrow Branch$

Then $S_ID \rightarrow Branch$

4. Union: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

5. Decomposition : If $X \rightarrow YZ$, then $X \rightarrow Y$ & $X \rightarrow Z$

6. Pseudo Transitive: If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$

7. Composition: If $X \rightarrow Y$ and $Z \rightarrow W$, then $XZ \rightarrow YW$

Closure Method:

This helps to find all candidate keys in a relation, where a candidate key is a minimal superkey.

Example:

1. R(ABCD) has FD $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

Now $A^+ = ABCD$

Similarly $B^+ = BCD$

$C^+ = CD$

$D^+ = D$

Closure of A contains all the attributes of the relation R and it is minimal so A is only candidate key.

2. R(ABCD) has FD $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

Now $A^+ = ABCD$

Similarly $B^+ = BCDA$

$C^+ = CDAB$

$D^+ = DABC$

Closure of A, B, C, and D contains all the attributes of the relation R and it is minimal so A, B, C, and D are all candidate keys.

3. R(ABCDE) has FD $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$

Here check if RHS contains all the attributes of the relation.
Since E is not determined by any FD,

$AE^+ = AEBCD$

Check BE, CE, DE